

Python ETL Jobs Assignment

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Date: 24th September, 2025

Step 1: Setting up Jupyter with VS Code

- **Create a New Clean Environment with Jupyter Support and Conda**

```
conda create -n <jupyter_env_name> python=3.10 ipykernel jupyter notebook -y
```

This will:

- Create a new env named <jupyter_env_name>
- Install Python 3.10
- Install jupyter and python extensions needed to run Jupyter notebooks from VS Code

- **Activate the New Environment**

```
conda activate <jupyter_env_name>
```

- **Register It with Jupyter**

VS Code usually detects this automatically, but to be safe, run:

```
python -m ipykernel install --user --name <jupyter_env_name> --display-name "Python (<jupyter_env_name>)"
```

This step ensures VS Code + Jupyter sees it as a valid kernel.

- **Open VS Code and Select the New Kernel**

1. Open VS Code
2. Open your .ipynb file
3. At the top-right of the notebook, click "**Select Kernel**"
4. Choose: Python (<jupyter_env_name>)

Step 2: Project Folder Structure

```
Etl_jobs/
├── data/
│   └── survey_results_public.csv  # raw dataset
├── notebooks/
│   ├── 1.ipynb
│   ├── 2.ipynb
│   ├── 3.ipynb
│   ├── 4.ipynb
│   └── 5.ipynb
```

```
├── 6.ipynb
├── test.ipynb      # tes notebook where we solve all 6 Qs
├── outputs/
│   ├── q2_python_pct_by_country.csv
│   ├── q3_salary_by_continent.csv
│   ├── q4_desired_lang_2020.csv
│   ├── q5_hobby_by_gender_continent.csv
│   └── q6_satisfaction_by_gender_continent.csv
└── README.md      # short doc explaining the project
```

Step 3: ETL Workflow (Extract → Transform → Load)

Extract

- Load dataset (`survey_results_public.csv`) into pandas.

Transform

Clean and prepare only the columns we need for each question:

1. **Q1 (Age1stCode)**
 - Column: `Age1stCode` (string like "12", "5-10", "Older than 40")
 - Convert to numeric (take midpoints for ranges, drop "Younger than 5" → 4, "Older than 40" → 41).
2. **Q2 (Python knowledge by country)**
 - Column: `LanguageWorkedWith` (semicolon-separated string).
 - Split into list, check if "Python" in it.
 - Group by `Country`, calculate percentage of developers who know Python.
3. **Q3 (Average salary by continent)**
 - Column: `ConvertedComp` (numeric salary in USD).
 - Column: `Country` → map to `Continent` (via `country_converter`).
 - Group by continent, take mean.
4. **Q4 (Most desired language for 2020)**
 - Column: `LanguageDesireNextYear` (semicolon-separated string).
 - Count frequency across all developers.
 - Pick the **most common** one.
5. **Q5 (Hobby by gender & continent)**
 - Column: `Hobbyist` ("Yes" or "No").
 - Column: `Gender` → Normalize into **MAN, WOMAN, OTHERS**:
 - If exactly "Man" → Man.
 - If exactly "Woman" → Woman.
 - Else → Others.
 - Map country → continent.
 - Group by (`Gender`, `Continent`), count hobbyist percentages.
6. **Q6 (Job & Career satisfaction by gender & continent)**
 - Columns: `JobSat`, `CareerSat`.
 - Normalize gender as in Q5.
 - Map country → continent.
 - Group by (`Gender`, `Continent`) → take average satisfaction.

Load

- Save results of each question as CSV into `outputs/`.
 - Also display them inside notebook with pretty tables.
-

Question 1:

```
# Find the average age of developers when they wrote their first line of code.
import pandas as pd
import numpy as np

# Load dataset
df = pd.read_csv('../data/survey_results_public.csv')

# Clean Age1stCode column
def clean_age(value):
    if pd.isna(value):
        return np.nan
    # Younger than 5 years
    if "Younger" in str(value):
        return 4
    # Older than 85
    if "Older" in str(value):
        return 86
    # Non-numeric values
    try:
        return int(value)
    except:
        return np.nan

# Apply cleaning function and drop NaN
df['Age1stCode_clean'] = df['Age1stCode'].apply(clean_age)
df = df.dropna(subset=['Age1stCode_clean'])

# Calculate average age
average_age = df['Age1stCode_clean'].mean()
print("Average age when developers wrote first line of code:", round(average_age, 2))
```

Output:

```
Average age when developers wrote first line of code: 15.41
```

Question 2:

```
# Deduce the percentage of developers who knew python in each country.

import pandas as pd

# Load dataset
df = pd.read_csv('../data/survey_results_public.csv')

# Drop NaN in LanguageWorkedWith and Country columns
df = df[['Country', 'LanguageWorkedWith']].dropna()

# Create a boolean column indicating if 'python' is in LanguageWorkedWith
df['KnowsPython'] = df['LanguageWorkedWith'].apply(lambda x: 'python' in
str(x).lower().split(';'))

# Group by Country and calculate total developers and python developers
result = df.groupby('Country').agg(
    Total_devs=('KnowsPython', 'count'),
    Python_devs=('KnowsPython', 'sum')
).reset_index()

# Calculate percentage of python developers
result['Python_percentage'] = (result['Python_devs'] / result['Total_devs']) * 100

# Sort by Python_percentage in descending order
result_sorted = result.sort_values('Python_percentage',
ascending=False).reset_index(drop=True)

# Save CSV
import os
os.makedirs('../output', exist_ok=True)
result_sorted.to_csv('../output/q2_developers_knew_python.csv', index=False)

# Display the top 10 countries with highest percentage of python developers
result_sorted.head(10)
```

Output:

	Country	Total_devs	Python_devs	Python_percentage
0	Dominica	1	1	100.000000
1	Niger	1	1	100.000000
2	Sierra Leone	1	1	100.000000
3	Sao Tome and Principe	1	1	100.000000
4	Timor-Leste	1	1	100.000000
5	Turkmenistan	7	6	85.714286
6	Mauritania	7	5	71.428571
7	Guinea	3	2	66.666667
8	Guyana	3	2	66.666667
9	Bahamas	3	2	66.666667

Question 3:

```
# Generate the report for the average salary of developers based on the continent.

import pandas as pd
import country_converter as coco

# Load dataset
df = pd.read_csv('../data/survey_results_public.csv')

# Keep only salary and country, drop missing values
df = df[['Country', 'ConvertedComp']].dropna()

# Exclude 'Other Country (Not Listed Above)'
df = df[df['Country'] != 'Other Country (Not Listed Above)']

# Map countries to continents
cc = coco.CountryConverter()
df['Continent'] = cc.convert(names=df['Country'], to='continent')

# Group by continent and calculate average salary
result = df.groupby('Continent').agg(
    Average_salary=('ConvertedComp', 'mean')
).reset_index()

# Sort by average salary descending
result_sorted = result.sort_values('Average_salary', ascending=False).reset_index(drop=True)

# Save CSV
import os
os.makedirs('../output', exist_ok=True)
result_sorted.to_csv('../output/q3_avg_salary_per_continent.csv', index=False)

# Display the top 10 continents by average salary
result_sorted.head(10)
```

Output:

	Continent	Average_salary
0	America	202277.425588
1	Oceania	166950.955791
2	Europe	97042.157518
3	Asia	37768.549543
4	Africa	35045.782816

Question 4:

Based on this survey, what will be the most desired programming language for the year 2020?

```
import pandas as pd

# Load dataset
df = pd.read_csv('../data/survey_results_public.csv')
df = df[['LanguageDesireNextYear']].dropna()

# Split and explode the languages
df['LangList'] = df['LanguageDesireNextYear'].astype(str).str.split(';')
df = df.explode('LangList')
df['LangList'] = df['LangList'].str.strip()

# Count occurrences of each language
lang_counts = df['LangList'].value_counts().reset_index()
lang_counts.columns = ['language', 'count']

# Save to CSV
import os
os.makedirs('../output', exist_ok=True)
lang_counts.to_csv('../output/q4_desired_lang_2020.csv', index=False)

# Display the top 10 desired languages
print("Top 10 desired languages for 2020:")
print(lang_counts.head(10))

# Identify the most desired language
most_desired = lang_counts.iloc[0]['language'] if not lang_counts.empty else None
print("\nMost desired language for 2020 (survey):", most_desired)
```

Output:

Top 10 desired languages for 2020:

	language	count
0	JavaScript	44739
1	Python	40006
2	HTML/CSS	36645
3	SQL	33566
4	TypeScript	23720
5	Java	23508
6	C#	22449
7	Bash/Shell/PowerShell	21303
8	Go	17060
9	C++	16856

Most desired language for 2020 (survey): JavaScript

Question 5:

```
# Generate the report for the people who code as a hobby based on their gender and continent.

import pandas as pd
import numpy as np
import country_converter as coco

# Load dataset
df = pd.read_csv('../data/survey_results_public.csv', low_memory=False)
df = df.dropna()

# Normalize Gender values
def normalize_gender(val):
    if pd.isna(val):
        return 'Others'
    s = str(val).strip()
    if s in ['Man', 'Woman']:
        return s
    else:
        return 'Others'

df['Gender'] = df['Gender'].apply(normalize_gender)

# Continent mapping
cc = coco.CountryConverter()
df = df[df['Country'] != 'Other Country (Not Listed Above)']
df['Continent'] = cc.convert(names=df['Country'], to='continent', not_found='Unknown')

# Create HobbyistFlag
df['HobbyistFlag'] = df['Hobbyist'].map(lambda x: True if str(x).strip() == 'Yes' else False)

# Group by Continent and Gender
result = df.groupby(['Continent', 'Gender']).agg(
    Total_respondents=('Respondent', 'nunique'),
    Hobbyists=('HobbyistFlag', 'sum')
).reset_index()

# Calculate percentage of hobbyists
result['pct_hobbyist'] = (result['Hobbyists'] / result['Total_respondents'] * 100).round(2)

# Save CSV
import os
os.makedirs('../output', exist_ok=True)
result.to_csv('../output/q5_code_as_hobby.csv', index=False)

# Display the first 10 rows of the result
result.head(10)
```

Output:

	Continent	Gender	Total_respondents	Hobbyists	pct_hobbyist
0	Africa	Man	65	56	86.15
1	Africa	Woman	4	2	50.00
2	America	Man	1399	1210	86.49
3	America	Others	11	11	100.00
4	America	Woman	58	40	68.97
5	Asia	Man	516	428	82.95
6	Asia	Others	2	2	100.00
7	Asia	Woman	13	7	53.85
8	Europe	Man	1246	1093	87.72
9	Europe	Others	1	1	100.00

Question 6:

```
# Generate the report for job and career satisfaction of developers based on their gender and
continent.

import pandas as pd
import numpy as np
import country_converter as coco

# Load dataset
df = pd.read_csv('../data/survey_results_public.csv', low_memory=False)

# Create a working dataframe
df = df[['Gender', 'Country', 'JobSat', 'CareerSat']]
df = df.dropna(subset=['Country', 'JobSat', 'CareerSat'])

# Normalize Gender values
def normalize_gender(val):
    if pd.isna(val):
        return 'Others'
    s = str(val).strip()
    if s in ['Man', 'Woman']:
        return s
    else:
        return 'Others'

df['Gender'] = df['Gender'].apply(normalize_gender)

# Country to Continent mapping
cc = coco.CountryConverter()
df = df[df['Country'] != 'Other Country (Not Listed Above)']
df['Continent'] = cc.convert(names=df['Country'], to='continent', not_found='Unknown')
```

```

# Mapping satisfaction to numeric
satisfaction_map = {
    'Very satisfied': 5,
    'Slightly satisfied': 4,
    'Neither satisfied nor dissatisfied': 3,
    'Slightly dissatisfied': 2,
    'Very dissatisfied': 1
}

df['JobSat'] = df['JobSat'].map(satisfaction_map)
df['CareerSat'] = df['CareerSat'].map(satisfaction_map)

# Group by Gender & Continent, calculate mean satisfaction
result = df.groupby(['Continent', 'Gender']).agg({
    'JobSat': 'mean',
    'CareerSat': 'mean',
}).round(2).reset_index()

# Save CSV
import os
os.makedirs('../output', exist_ok=True)
result.to_csv('../output/q6_job_carrer_satisfaction.csv')

# Display top 10 rows
result.head(10)

```

Output:

		JobSat	CareerSat
Continent	Gender		
Africa	Man	3.46	3.79
	Others	3.21	3.26
	Woman	3.23	3.69
America	Man	3.82	4.13
	Others	3.65	3.92
	Woman	3.94	4.17
Asia	Man	3.43	3.70
	Others	3.40	3.63
	Woman	3.46	3.67
Europe	Man	3.70	3.94

Summary:

We loaded the **2019 StackOverflow survey** dataset using pandas, handled missing values, and explored key columns like **Age1stCode**, **LanguageWorkedWith**, **Country**, **ConvertedComp**, **Gender**, **Hobbyist**, **JobSat**, and **CareerSat**.

- **Q1:** Calculated the average age developers wrote their first line of code.
- **Q2:** Determined the percentage of developers knowing Python per country.
- **Q3:** Converted countries to continents and computed average developer salaries by continent.
- **Q4:** Identified the most desired programming languages for 2020.
- **Q5:** Normalized gender and analyzed hobbyist coding trends by gender and continent.
- **Q6:** Mapped job and career satisfaction to numeric values and aggregated by gender and continent.

All results were exported as CSV reports.
